Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C/C++, SQL, JavaScript

Software Engineering: Data Structures & Algorithms, OOP, REST APIs, Multithreading, Backend Development, Unit Testing

AI & Machine Learning: Machine Learning, Deep Learning, LLMs, RAG, Prompt Engineering, PyTorch, TensorFlow/Keras

Web & Backend: FastAPI, Flask, React.js, Node.js, Streamlit

Databases & Caching: SQL, SQLite, MySQL, MongoDB, Redis

Cloud & Deployment: AWS, Docker, GitHub, CI/CD

Data & Visualization: Pandas, NumPy, Power BI, Matplotlib, Plotly

Engineering Practices: Git/GitHub, PyTest, TDD, Debugging, Agile/Scrum, Reproducible Development

EXPERIENCE

AI for Sustainability Virtual Intern **|LINK** — ***July – Sep, 2026***

- Completed an AI-focused virtual internship organized by 1M1B covering Generative AI, Prompt Engineering, RAG, Agentic AI, and Responsible AI, with hands-on exposure to enterprise AI workflows and LLM applications.
- Applied RAG and semantic search concepts using document ingestion, embeddings, vector databases, and retrieval pipelines to improve the grounding and reliability of LLM responses.
- Built an Energy Anomaly Detector using Python, Pandas, NumPy, and Isolation Forest, detecting 9 anomalies across 6,918 test hours (0.13%), with automated email alerts, privacy safeguards, transparent scoring, and rate-limited notifications.

PROJECTS

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

Docwise **|LINK**

- Engineered an end-to-end RAG-based AI document pipeline using Google Gemini, NLP, TF-IDF retrieval, and intelligent chunking, achieving 95%+ query accuracy on PDFs exceeding 200 pages.
- Developed automated document-processing workflows covering extraction, chunking, retrieval, contextual ranking, and LLM-based response generation.
- Improved AI output reliability and validation through retrieval tuning, prompt engineering, and hallucination-aware checks, increasing contextual relevance and answer consistency.

API Rate Limiter Library **|LINK**

- Built a Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for API automation and reliable traffic control.
- Designed a modular, thread-safe architecture with pluggable storage backends to support concurrent request handling and reliable API behavior across application environments.
- Developed 30+ PyTest automated tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for reusable software development.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — **2026**